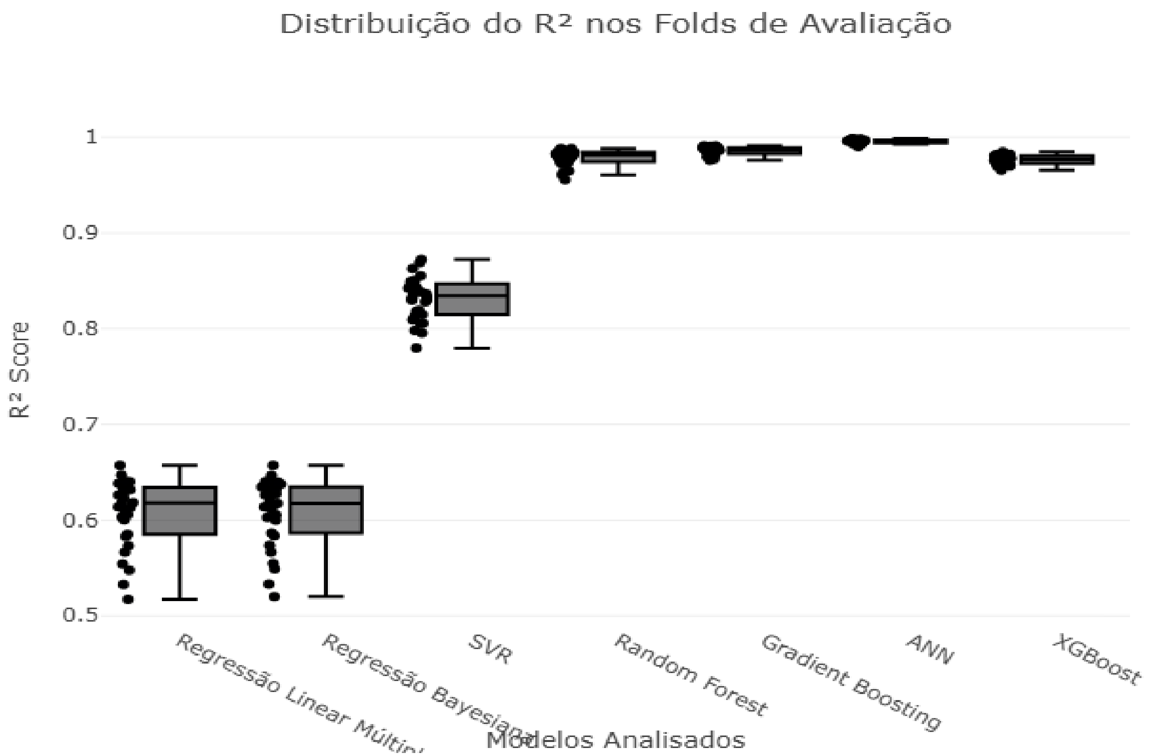
Relatório Técnico de Modelagem e Auditoria (Regressão)

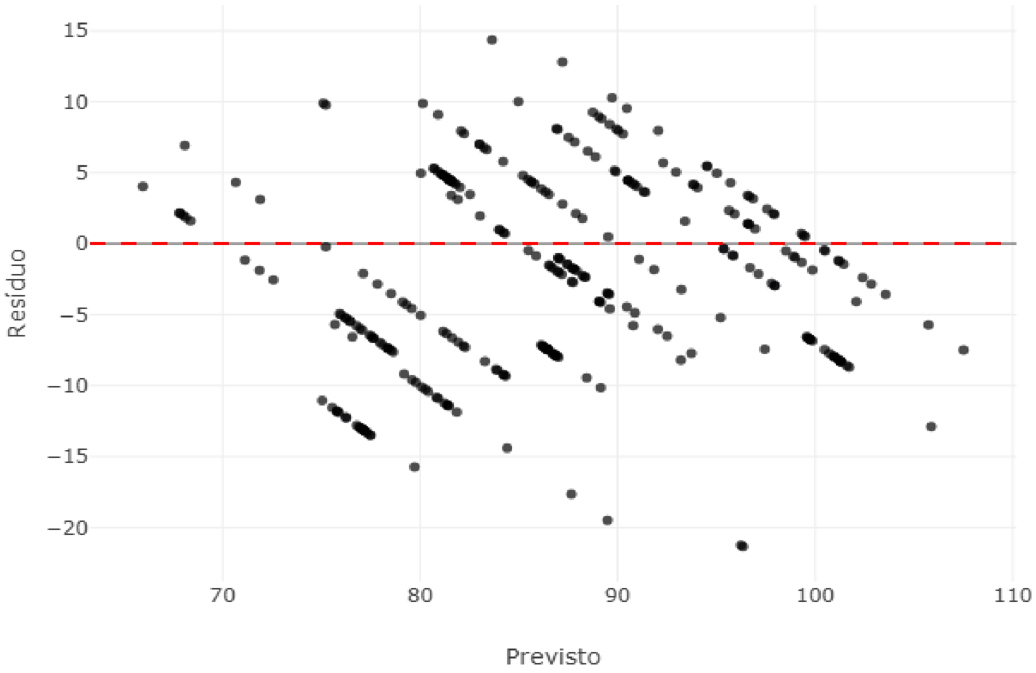
1. Desempenho Global

do	R2 (CV)	MAE (CV)	RMSE (CV)	R2 (Externo)	MAE (Externo)	RMSE (Externo)	Configuração
g	0.9865587493480042	0.5813518174609429	1.216184272102395	0.8518644360730382	2.708447091092474	4.2494330817571795	{'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 400, 'subsam
	0.9745584129256354	0.8387396105770694	1.6734335454918032	0.8316509408788435	2.6611815096914966	4.530088367120988	{'colsample_bytree': 0.6, 'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 5
	0.9941499241450407	0.3482578797291491	0.7335631468245475	0.8002397998177384	2.8418608667363796	4.934643800457617	{'activation': 'tanh', 'alpha': 0.1, 'hidden_layer_sizes': (100, 100), 'sol
t	0.9810641575971882	0.6688569660555571	1.4494922038570597	0.7341217589224163	3.41351431065007	5.693020209514571	{'max_depth': 10, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_est
ana	0.6050720107446926	5.244135400785949	6.655273062970611	0.5930361003134813	5.905172123498843	7.043350001596069	{'alpha_1': 1.0, 'lambda_1': 1e-06}
últipla	0.6050123935692086	5.242761720801656	6.6555469840699795	0.5921217597052069	5.921600347180739	7.051257837807291	{'fit_intercept': True}
	0.8340068614848796	3.0730230437656645	4.30367131872429	0.4965629308026035	6.104902366077604	7.833826021274597	{'C': 10, 'epsilon': 0.5, 'kernel': 'rbf'}

Análise Estatística: Estabilidade (Teste de Friedman)



Diagnóstico Analítico: Regressão Linear Múltipla

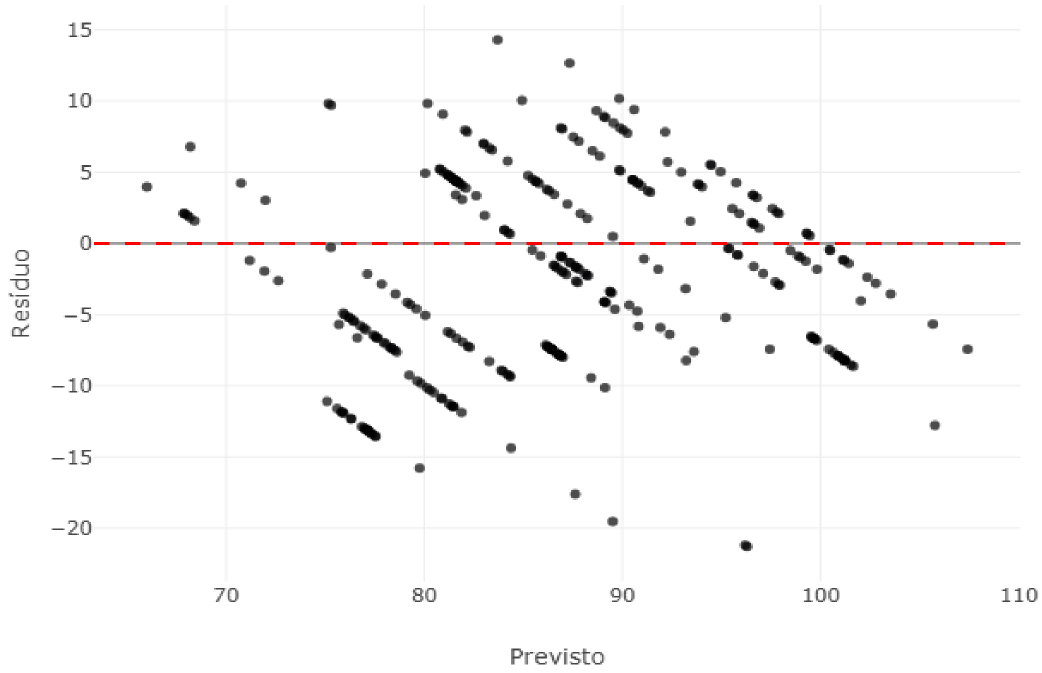


Amostra de Erros Críticos (Piores Predições)

TeorRotulo	TeorDensimetro	NivelAcucar	Temperatura	CPJ_10k	CPJ_4k7	CPJ_3k19	CPJ_1k	CPJ_824R	Real	Previsto	Resíduo	Erro Absoluto	Erro Relativo (%)
39	34	15	21.5	16360.9	26214.3	26263.6	28945.6	29238.3	75.0	96.33743977451064	-21.33743977451064..	21.337439774510642	28.44991969934752
39	34	15	21.5	16421.1	26190.2	26235.9	28859.9	29129.8	75.0	96.23451951795766	-21.23451951795766..	21.234519517957665	28.31269269061022
42	39	0	25.9	15845.8	25860.5	25928.5	28662.7	29036.8	70.0	89.48340305653387	-19.48340305653387..	19.483403056533874	27.833432937905535
42	39	0	19.1	17281.4	26207.0	26384.7	29150.9	29405.9	70.0	87.65287071067567	-17.65287071067567	17.65287071067567	25.218386729536675
38	41	0	20.7	20349.4	24717.9	24966.6	27101.7	27329.0	64.0	79.71885885743883	-15.71885885743883..	15.718858857438832	24.560716964748174
42	39	0	19.1	17548.1	26107.9	26274.6	29101.2	29374.4	70.0	84.40262150128511	-14.40262150128511..	14.402621501285111	20.575173573264443
39	34	15	19.1	17383.0	25700.9	25737.1	28238.0	28495.6	98.0	83.63995632264464	14.360043677355364	14.360043677355364	14.653105793219758
38	41	0	21.4	20378.1	24736.7	24991.8	27170.0	27363.4	64.0	77.49332332921003	-13.49332332921002..	13.493323329210028	21.08331770189067

38	41	0	21.4	20198.4	24756.7	25012.0	27240.2	27432.7	64.0	77.46651478666456	-13.46651478666456..	13.466514786664561	21.041429354163377
38	41	0	21.4	20218.4	24753.7	25009.3	27241.5	27436.7	64.0	77.31177826738497	-13.31177826738496..	13.311778267384966	20.79965354278901
38	41	0	21.4	20340.9	24738.9	24993.9	27183.3	27370.3	64.0	77.24469700939318	-13.24469700939317..	13.244697009393178	20.69483907717684
38	41	0	21.4	20310.8	24736.3	24993.7	27189.5	27385.9	64.0	77.17836791205252	-13.17836791205252..	13.178367912052522	20.591199862582066
38	41	0	21.4	20265.1	24746.2	25002.5	27221.6	27415.5	64.0	77.14431869115283	-13.14431869115283..	13.144318691152833	20.5379979549263
38	41	0	21.4	20277.4	24740.1	24996.0	27207.6	27400.8	64.0	77.09141114248833	-13.09141114248832..	13.091411142488326	20.45532991013801
38	41	0	21.4	20363.4	24737.3	24994.3	27177.8	27363.5	64.0	77.0591229174839	-13.05912291748390..	13.059122917483904	20.404879558568602
38	41	0	21.4	20313.8	24740.3	24997.7	27194.1	27382.5	64.0	77.04965136098123	-13.04965136098123	13.04965136098123	20.39008025153317
38	41	0	21.4	20256.8	24743.7	25000.0	27228.1	27424.1	64.0	76.97464739437514	-12.97464739437514	12.97464739437514	20.272886553711157
38	41	0	20.7	20429.5	24725.4	24982.1	27168.1	27352.8	64.0	76.90809719036824	-12.90809719036823..	12.908097190368238	20.168901859950374
38	41	0	25.2	17525.5	25457.9	25633.8	28263.6	28628.3	93.0	105.87295276684077	-12.87295276684076..	12.872952766840768	13.841884695527709
38	41	0	18.3	18796.0	24698.2	24936.4	26865.5	27048.8	100.0	87.20232938838842	12.79767061161158	12.79767061161158	12.797670611611577

Diagnóstico Analítico: Regressão Bayesiana

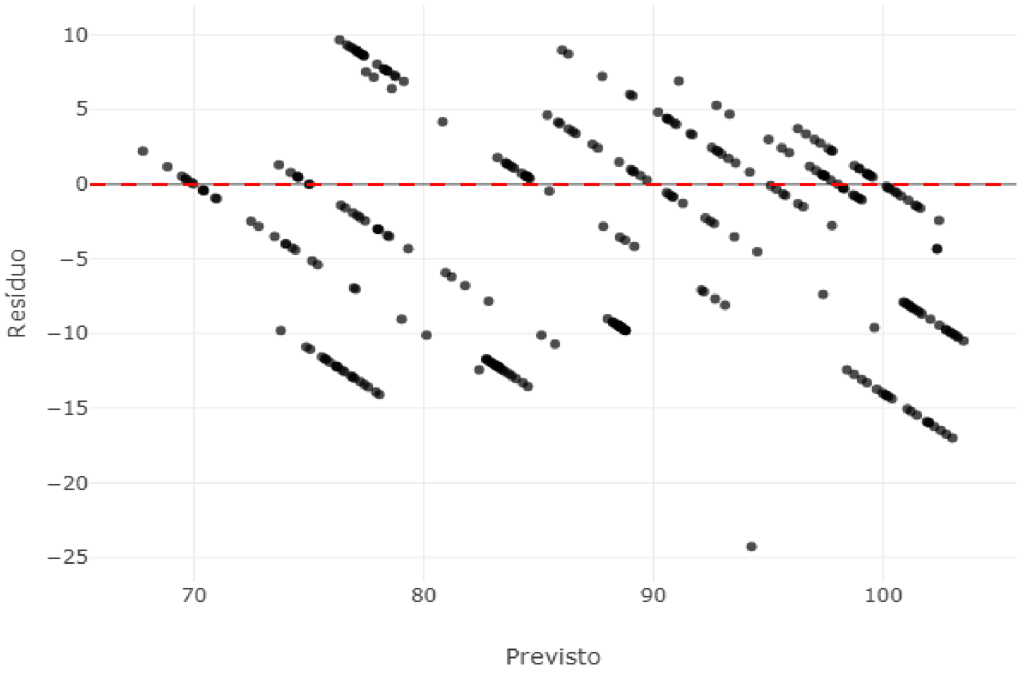


Amostra de Erros Críticos (Piores Predições)

TeorRotulo	TeorDensimetro	NivelAcucar	Temperatura	CPJ_10k	CPJ_4k7	CPJ_3k19	CPJ_1k	CPJ_824R	Real	Previsto	Resíduo	Erro Absoluto	Erro Relativo (%)
39	34	15	21.5	16360.9	26214.3	26263.6	28945.6	29238.3	75.0	96.28574483510698	-21.28574483510698..	21.285744835106982	28.380993113475977
39	34	15	21.5	16421.1	26190.2	26235.9	28859.9	29129.8	75.0	96.18722364580475	-21.18722364580475	21.18722364580475	28.249631527739666
42	39	0	25.9	15845.8	25860.5	25928.5	28662.7	29036.8	70.0	89.5069928595212	-19.50699285952120..	19.506992859521205	27.867132656458864
42	39	0	19.1	17281.4	26207.0	26384.7	29150.9	29405.9	70.0	87.6168774874279	-17.6168774874279	17.6168774874279	25.166967839182714
38	41	0	20.7	20349.4	24717.9	24966.6	27101.7	27329.0	64.0	79.76308444516813	-15.76308444516813..	15.763084445168133	24.629819445575208
42	39	0	19.1	17548.1	26107.9	26274.6	29101.2	29374.4	70.0	84.37773688743707	-14.37773688743706..	14.377736887437067	20.539624124910095
39	34	15	19.1	17383.0	25700.9	25737.1	28238.0	28495.6	98.0	83.69471507196259	14.305284928037409	14.305284928037409	14.59722951840552
38	41	0	21.4	20378.1	24736.7	24991.8	27170.0	27363.4	64.0	77.54230291411666	-13.54230291411666	13.54230291411666	21.15984830330728

38	41	0	21.4	20198.4	24756.7	25012.0	27240.2	27432.7	64.0	77.52164539861111	-13.52164539861111..	13.521645398611113	21.127570935329864
38	41	0	21.4	20218.4	24753.7	25009.3	27241.5	27436.7	64.0	77.3666136183392	-13.36661361833920..	13.366613618339201	20.885333778655003
38	41	0	21.4	20340.9	24738.9	24993.9	27183.3	27370.3	64.0	77.29705944313768	-13.29705944313768..	13.297059443137684	20.776655379902632
38	41	0	21.4	20310.8	24736.3	24993.7	27189.5	27385.9	64.0	77.23231457127	-13.23231457126999..	13.232314571269995	20.675491517609366
38	41	0	21.4	20265.1	24746.2	25002.5	27221.6	27415.5	64.0	77.19902095300488	-13.19902095300487..	13.199020953004876	20.62347023907012
38	41	0	21.4	20277.4	24740.1	24996.0	27207.6	27400.8	64.0	77.14685224896618	-13.14685224896618..	13.146852248966184	20.541956639009662
38	41	0	21.4	20363.4	24737.3	24994.3	27177.8	27363.5	64.0	77.11186821875307	-13.11186821875307..	13.111868218753074	20.48729409180168
38	41	0	21.4	20313.8	24740.3	24997.7	27194.1	27382.5	64.0	77.10425341868205	-13.10425341868204..	13.104253418682049	20.4753959666907
38	41	0	21.4	20256.8	24743.7	25000.0	27228.1	27424.1	64.0	77.03058172229623	-13.03058172229623..	13.030581722296233	20.360283941087864
38	41	0	20.7	20429.5	24725.4	24982.1	27168.1	27352.8	64.0	76.96424015788338	-12.96424015788338..	12.964240157883381	20.256625246692785
38	41	0	22.3	20176.4	24756.8	25011.9	27258.0	27454.3	64.0	76.84350704603308	-12.84350704603308..	12.843507046033082	20.06797975942669
38	41	0	25.2	17525.5	25457.9	25633.8	28263.6	28628.3	93.0	105.7707056792872	-12.77070567928720..	12.770705679287204	13.731941590631402

Diagnóstico Analítico: SVR

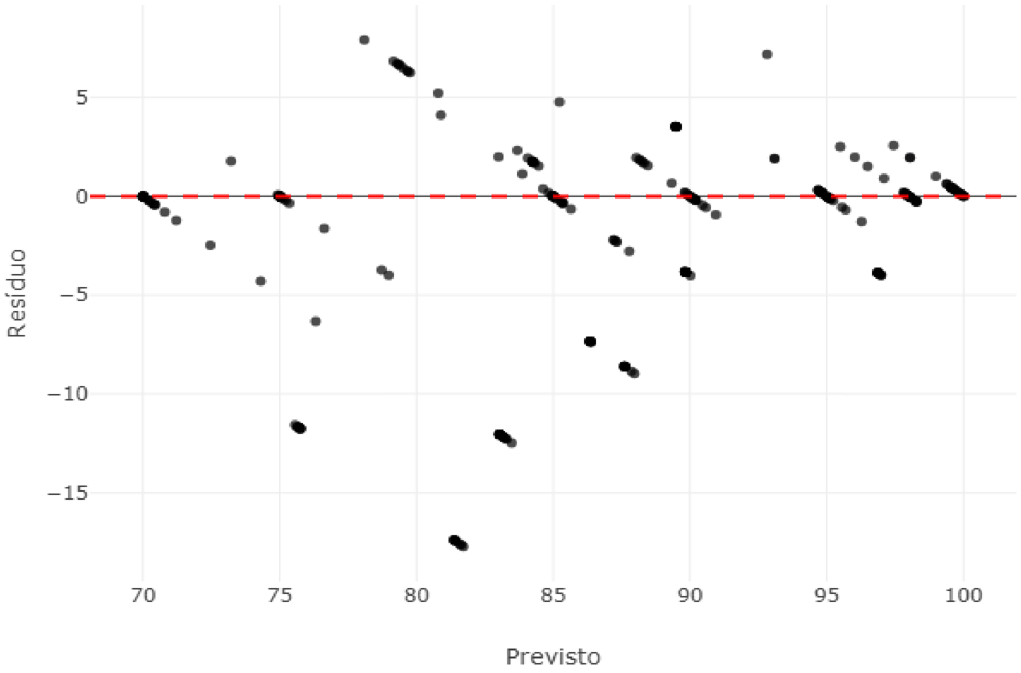


Amostra de Erros Críticos (Piores Predições)

TeorRotulo	TeorDensimetro	NivelAcucar	Temperatura	CPJ_10k	CPJ_4k7	CPJ_3k19	CPJ_1k	CPJ_824R	Real	Previsto	Resíduo	Erro Absoluto	Erro Relativo (%)
42	39	0	25.9	15845.8	25860.5	25928.5	28662.7	29036.8	70.0	94.26734331678468	-24.26734331678467..	24.267343316784675	34.66763330969239
42	39	0	23.4	16627.6	25335.8	25351.2	27849.2	28193.4	86.0	103.01080052716826	-17.01080052716825..	17.010800527168257	19.780000612986345
42	39	0	23.4	16665.7	25347.4	25363.8	27868.3	28217.9	86.0	102.74364651454042	-16.74364651454041..	16.743646514540416	19.469356412256296
42	39	0	23.4	16694.9	25357.7	25377.3	27890.2	28239.5	86.0	102.50526078417313	-16.50526078417313	16.50526078417313	19.192163702526894
42	39	0	23.4	16732.7	25368.3	25389.3	27906.2	28256.7	86.0	102.21985125182698	-16.21985125182698	16.21985125182698	18.860292153287187
42	39	0	23.4	16755.8	25381.6	25408.3	27924.0	28277.4	86.0	101.99567168115863	-15.99567168115862..	15.995671681158626	18.599618233905378
42	39	0	23.4	16764.8	25376.7	25400.9	27919.7	28271.7	86.0	101.96371959313069	-15.96371959313069	15.96371959313069	18.562464643175222
42	39	0	23.4	16766.4	25394.8	25424.2	27926.0	28297.1	86.0	101.89231086986723	-15.89231086986723	15.89231086986723	18.479431244031662

42	39	0	23.4	16814.2	25389.1	25417.2	27970.8	28333.4	86.0	101.46807506011012	-15.46807506011012..	15.468075060110124	17.986133790825725
42	39	0	23.4	16837.8	25404.4	25437.6	27987.3	28352.1	86.0	101.21069494287171	-15.21069494287171..	15.210694942871712	17.686854584734547
42	39	0	23.4	16860.2	25407.4	25436.6	27991.0	28354.8	86.0	101.05503318611858	-15.05503318611857..	15.055033186118578	17.505852541998344
42	39	0	22.5	16907.8	25416.5	25448.9	28012.7	28379.1	86.0	100.38715729400352	-14.38715729400351..	14.387157294003515	16.729252667445948
42	39	0	22.5	16928.0	25412.0	25448.2	28012.6	28380.0	86.0	100.21989280922361	-14.21989280922360..	14.219892809223609	16.534759080492567
42	39	0	22.5	16929.8	25420.1	25455.5	28025.5	28391.1	86.0	100.14400270348409	-14.14400270348409..	14.144002703484091	16.446514771493128
42	39	0	22.5	16947.0	25409.4	25445.7	28010.7	28375.3	86.0	100.07991426847639	-14.07991426847638..	14.079914268476386	16.371993335437658
38	41	0	20.7	20349.4	24717.9	24966.6	27101.7	27329.0	64.0	78.07192483307149	-14.07192483307149..	14.071924833071492	21.987382551674205
42	39	0	22.5	16963.6	25409.6	25443.0	28008.1	28373.5	86.0	99.96674430422583	-13.96674430422582..	13.966744304225827	16.24040035375096
38	41	0	20.7	20429.5	24725.4	24982.1	27168.1	27352.8	64.0	77.91833511606085	-13.91833511606084..	13.918335116060845	21.74739861884507
42	39	0	22.5	16986.9	25413.4	25451.1	28015.4	28380.3	86.0	99.7257146773515	-13.72571467735150..	13.725714677351505	15.960133345757562
38	41	0	21.4	20198.4	24756.7	25012.0	27240.2	27432.7	64.0	77.58387403689754	-13.58387403689754	13.58387403689754	21.22480318265241

Diagnóstico Analítico: Random Forest

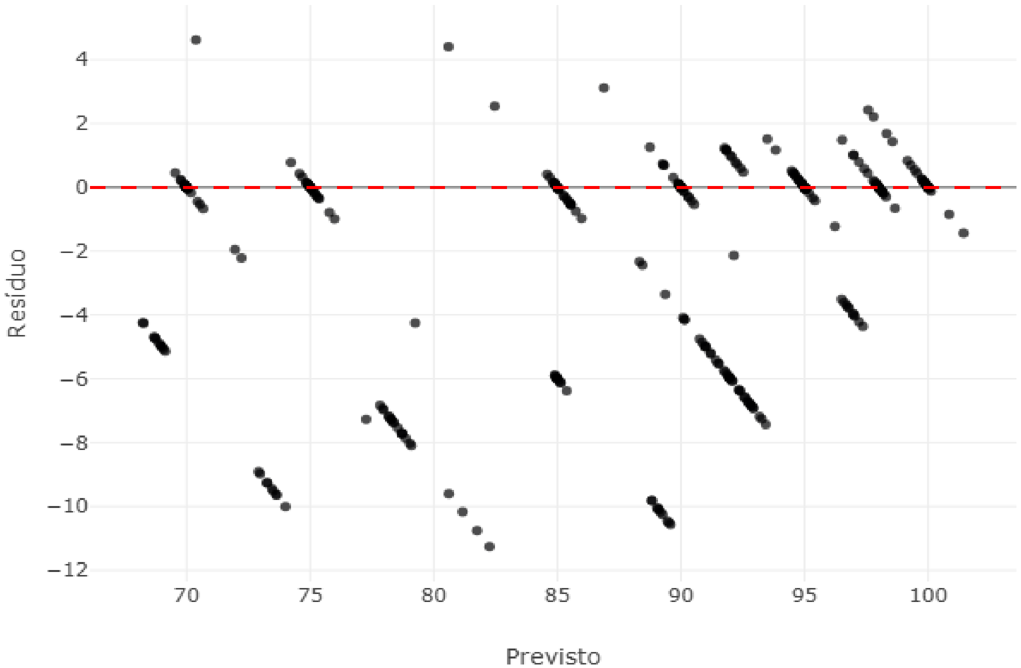


Amostra de Erros Críticos (Piores Predições)

TeorRotulo	TeorDensimetro	NivelAcucar	Temperatura	CPJ_10k	CPJ_4k7	CPJ_3k19	CPJ_1k	CPJ_824R	Real	Previsto	Resíduo	Erro Absoluto	Erro Relativo (%)
38	41	0	20.7	20349.4	24717.9	24966.6	27101.7	27329.0	64.0	81.71617660509077	-17.71617660509076..	17.716176605090766	27.68152594545432
38	41	0	20.7	20669.8	24697.2	24953.7	27142.7	27336.6	64.0	81.62824440008025	-17.62824440008025	17.62824440008025	27.544131875125387
38	41	0	20.7	20429.5	24725.4	24982.1	27168.1	27352.8	64.0	81.6008532220062	-17.60085322200619..	17.600853222006194	27.50133315938468
38	41	0	20.7	20514.8	24714.2	24971.8	27175.4	27368.9	64.0	81.47042101699567	-17.47042101699567..	17.470421016995672	27.297532839055737
38	41	0	20.7	20550.6	24714.4	24971.8	27172.3	27366.2	64.0	81.42491106674692	-17.42491106674691..	17.424911066746915	27.226423541792055
38	41	0	20.7	20865.9	24671.3	24926.8	27092.4	27283.0	64.0	81.39343219350748	-17.39343219350748	17.39343219350748	27.177237802355435
38	41	0	20.7	20592.2	24703.3	24961.7	27156.8	27351.4	64.0	81.38602217785802	-17.38602217785802..	17.386022177858024	27.165659652903162
38	41	0	20.7	20735.2	24687.6	24944.3	27122.2	27315.7	64.0	81.38491106674691	-17.38491106674691	17.38491106674691	27.163923541792045

38	41	0	20.7	20629.8	24704.8	24964.0	27157.8	27352.0	64.0	81.37491106674692	-17.37491106674691..	17.374911066746918	27.14829854179206
42	39	0	26.0	17408.6	25889.7	26003.5	29025.1	29428.2	71.0	83.48333333333333	-12.48333333333333..	12.483333333333334	17.582159624413148
42	39	0	26.0	17355.9	25906.9	26027.0	29062.6	29462.8	71.0	83.3	-12.29999999999999..	12.299999999999997	17.32394366197183
42	39	0	26.0	17326.8	25919.2	26047.3	29099.1	29503.4	71.0	83.25	-12.25	12.25	17.253521126760564
42	39	0	26.0	17303.8	25936.5	26061.8	29121.0	29522.1	71.0	83.23333333333333	-12.23333333333333..	12.233333333333334	17.230046948356808
42	39	0	25.7	17414.3	25966.3	26120.0	29205.4	29586.2	71.0	83.18333333333334	-12.18333333333333..	12.183333333333337	17.159624413145544
42	39	0	25.7	17399.1	25968.4	26120.0	29208.2	29591.9	71.0	83.15	-12.15000000000000..	12.150000000000006	17.112676056338035
42	39	0	26.0	17288.9	25943.9	26072.4	29138.5	29538.4	71.0	83.15	-12.15000000000000..	12.150000000000006	17.112676056338035
42	39	0	26.0	17283.7	25952.5	26082.7	29154.7	29553.7	71.0	83.15	-12.15000000000000..	12.150000000000006	17.112676056338035
42	39	0	25.7	17405.5	25969.4	26123.6	29208.3	29589.2	71.0	83.13333333333334	-12.13333333333334	12.13333333333334	17.089201877934283
42	39	0	25.7	17354.5	25960.7	26103.8	29188.9	29580.4	71.0	83.06666666666666	-12.06666666666666..	12.066666666666663	16.995305164319245
42	39	0	25.7	17336.5	25964.0	26103.4	29187.7	29579.4	71.0	83.06666666666666	-12.06666666666666..	12.066666666666663	16.995305164319245

Diagnóstico Analítico: Gradient Boosting

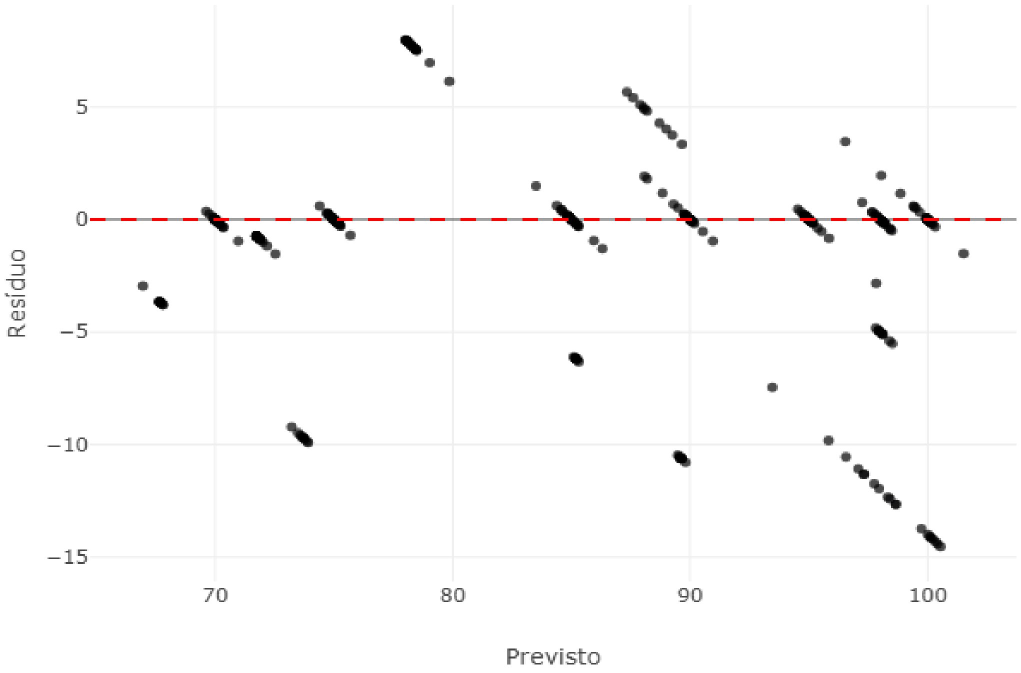


Amostra de Erros Críticos (Piores Predições)

TeorRotulo	TeorDensimetro	NivelAcucar	Temperatura	CPJ_10k	CPJ_4k7	CPJ_3k19	CPJ_1k	CPJ_824R	Real	Previsto	Resíduo	Erro Absoluto	Erro Relativo (%)
42	39	0	26.0	17326.8	25919.2	26047.3	29099.1	29503.4	71.0	82.25617217379042	-11.25617217379041..	11.256172173790418	15.853763625056928
42	39	0	26.0	17408.6	25889.7	26003.5	29025.1	29428.2	71.0	81.75247263864125	-10.75247263864125..	10.752472638641251	15.1443276600581
38	41	0	25.3	18924.7	25136.2	25392.3	27721.3	27893.1	79.0	89.57259731388778	-10.57259731388778..	10.572597313887783	13.383034574541497
38	41	0	25.3	18931.5	25138.6	25393.2	27722.5	27893.9	79.0	89.50730985219528	-10.50730985219527..	10.507309852195277	13.300392217968707
38	41	0	25.3	18937.0	25134.0	25389.1	27713.4	27886.4	79.0	89.45348547668229	-10.45348547668228..	10.453485476682289	13.23226009706619
38	41	0	25.3	18917.1	25139.7	25394.3	27725.2	27898.9	79.0	89.25542334847727	-10.25542334847726..	10.255423348477265	12.981548542376284
38	41	0	25.3	18913.1	25143.2	25389.7	27718.3	27912.8	79.0	89.18160171243856	-10.18160171243856..	10.181601712438564	12.888103433466537
42	39	0	26.0	17355.9	25906.9	26027.0	29062.6	29462.8	71.0	81.16520145276836	-10.16520145276835..	10.165201452768358	14.317185144744165

38	41	0	25.3	18910.7	25141.9	25397.6	27731.1	27903.0	79.0	89.1073936630221	-10.10739366302209..	10.107393663022094	12.794169193698854
38	41	0	25.3	18934.6	25148.1	25401.0	27740.2	27897.4	79.0	89.07139940646287	-10.07139940646287..	10.071399406462874	12.748606843623891
38	41	0	25.3	18910.2	25144.2	25400.9	27735.2	27885.0	79.0	89.04123673517029	-10.04123673517028..	10.041236735170287	12.710426247050997
38	41	0	20.7	20865.9	24671.3	24926.8	27092.4	27283.0	64.0	73.99752098419033	-9.99752098419033	9.99752098419033	15.621126537797391
38	41	0	25.3	18920.3	25146.3	25401.3	27739.9	27895.0	79.0	88.81951290274486	-9.819512902744862	9.819512902744862	12.429763168031469
38	41	0	25.3	18911.7	25144.4	25403.3	27738.0	27893.7	79.0	88.81951290274486	-9.819512902744862	9.819512902744862	12.429763168031469
38	41	0	20.7	20735.2	24687.6	24944.3	27122.2	27315.7	64.0	73.64198725621223	-9.641987256212232	9.641987256212232	15.065605087831614
38	41	0	20.7	20669.8	24697.2	24953.7	27142.7	27336.6	64.0	73.61972902435626	-9.61972902435626	9.61972902435626	15.030826600556658
42	39	0	26.0	17303.8	25936.5	26061.8	29121.0	29522.1	71.0	80.6067096540985	-9.606709654098495	9.606709654098495	13.530576977603515
38	41	0	20.7	20349.4	24717.9	24966.6	27101.7	27329.0	64.0	73.50675696919494	-9.506756969194939	9.506756969194939	14.854307764367093
38	41	0	20.7	20429.5	24725.4	24982.1	27168.1	27352.8	64.0	73.44187050409286	-9.441870504092861	9.441870504092861	14.752922662645096
38	41	0	20.7	20629.8	24704.8	24964.0	27157.8	27352.0	64.0	73.26328045581505	-9.26328045581505	9.26328045581505	14.473875712211015

Diagnóstico Analítico: ANN

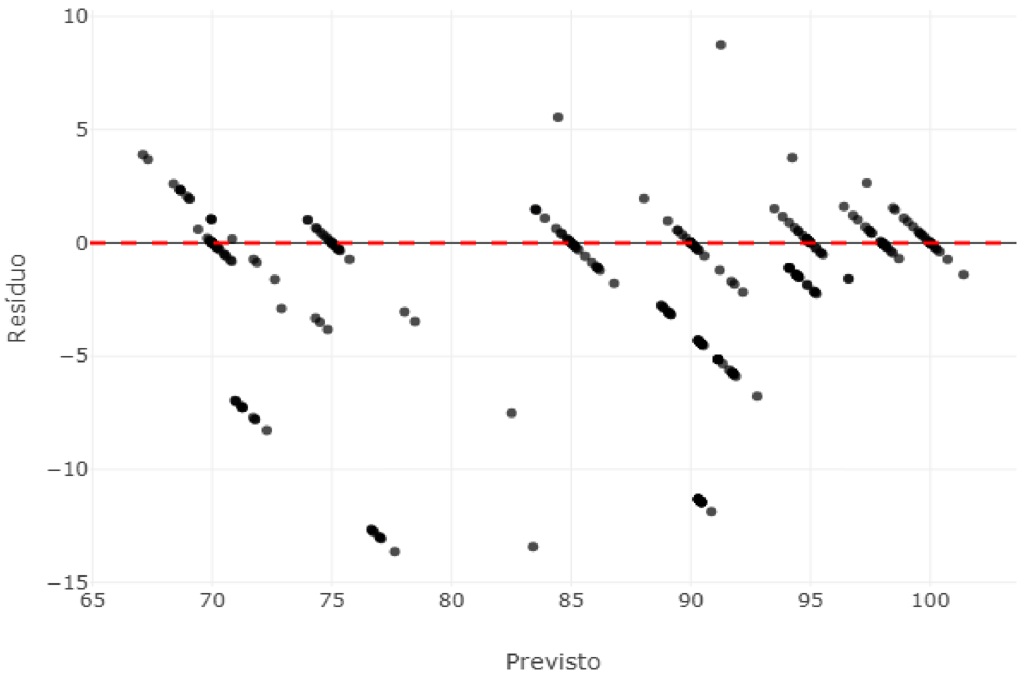


Amostra de Erros Críticos (Piores Predições)

TeorRotulo	TeorDensimetro	NivelAcucar	Temperatura	CPJ_10k	CPJ_4k7	CPJ_3k19	CPJ_1k	CPJ_824R	Real	Previsto	Resíduo	Erro Absoluto	Erro Relativo (%)
39	38	5	26.0	20587.6	24275.5	24512.5	26414.1	26597.9	86.0	100.53855606843126	-14.53855606843126..	14.538556068431262	16.90529775398984
39	38	5	26.0	20655.0	24261.9	24491.4	26425.5	26585.7	86.0	100.41281662115216	-14.41281662115216	14.41281662115216	16.759089094362977
39	38	5	26.0	20729.5	24255.4	24498.1	26441.0	26586.4	86.0	100.3265167491258	-14.32651674912580..	14.326516749125801	16.658740405960234
39	38	5	26.0	20758.5	24244.4	24483.1	26394.1	26575.0	86.0	100.20504112247139	-14.20504112247138..	14.205041122471386	16.517489677292307
39	38	5	26.0	20766.1	24237.4	24477.0	26401.0	26560.5	86.0	100.13185230252446	-14.13185230252446..	14.131852302524464	16.432386398284258
39	38	5	26.0	20783.0	24233.4	24476.5	26467.1	26637.6	86.0	100.0896074011364	-14.08960740113640..	14.089607401136405	16.383264419926054
39	38	5	26.0	20779.0	24226.7	24468.1	26448.8	26616.2	86.0	100.00942397024627	-14.00942397024627..	14.009423970246274	16.29002787237939
39	38	5	26.0	20797.7	24214.9	24452.7	26430.3	26600.7	86.0	99.73327640122409	-13.73327640122408..	13.733276401224089	15.968926047934987

39	38	5	26.2	20848.7	24181.1	24420.3	26382.2	26547.3	86.0	98.65854991715418	-12.65854991715417..	12.658549917154176	14.719244089714156
39	38	5	26.2	20894.0	24182.3	24418.5	26394.6	26561.2	86.0	98.65654960545244	-12.65654960545244	12.65654960545244	14.71691814587493
39	38	5	26.2	20870.1	24175.2	24415.7	26387.0	26554.6	86.0	98.41550311744851	-12.41550311744850..	12.415503117448509	14.436631531916872
39	38	5	26.2	20916.1	24175.0	24412.9	26392.5	26557.3	86.0	98.32755985575247	-12.32755985575246..	12.327559855752469	14.334371925293569
39	38	5	26.2	20935.3	24169.3	24406.7	26383.9	26551.3	86.0	97.95914969122111	-11.95914969122111..	11.959149691221114	13.905988013047807
39	38	5	26.2	20976.2	24166.2	24403.1	26387.9	26554.6	86.0	97.74966215602808	-11.74966215602808	11.74966215602808	13.662397855846606
39	38	5	26.0	20805.8	24171.0	24412.9	26366.3	26531.9	86.0	97.31665556481214	-11.31665556481213..	11.316655564812137	13.158901819548996
39	38	5	26.0	20823.6	24168.5	24413.3	26373.1	26537.8	86.0	97.31336817656857	-11.31336817656857..	11.313368176568574	13.155079275079737
39	38	5	26.2	20993.4	24159.1	24393.6	26377.4	26545.8	86.0	97.0788574525566	-11.07885745255660..	11.078857452556605	12.882392386693727
38	41	0	25.3	18913.1	25143.2	25389.7	27718.3	27912.8	79.0	89.80717914218761	-10.80717914218760..	10.807179142187607	13.679973597705834
38	41	0	25.3	18934.6	25148.1	25401.0	27740.2	27897.4	79.0	89.64494400109473	-10.64494400109472..	10.644944001094728	13.47461265961358
38	41	0	25.3	18910.2	25144.2	25400.9	27735.2	27885.0	79.0	89.63174263654477	-10.63174263654477..	10.631742636544772	13.457902071575662

Diagnóstico Analítico: XGBoost



Amostra de Erros Críticos (Piores Predições)

TeorRotulo	TeorDensimetro	NivelAcucar	Temperatura	CPJ_10k	CPJ_4k7	CPJ_3k19	CPJ_1k	CPJ_824R	Real	Previsto	Resíduo	Erro Absoluto	Erro Relativo (%)
38	41	0	20.7	20735.2	24687.6	24944.3	27122.2	27315.7	64.0	77.62737	-13.62737274169921..	13.627372741699219	21.29276990890503
42	39	0	19.1	17548.1	26107.9	26274.6	29101.2	29374.4	70.0	83.406746	-13.40674591064453..	13.406745910644531	19.152494158063615
38	41	0	20.7	20669.8	24697.2	24953.7	27142.7	27336.6	64.0	77.06895	-13.06894683837890..	13.068946838378906	20.42022943496704
38	41	0	20.7	20865.9	24671.3	24926.8	27092.4	27283.0	64.0	77.02678	-13.02677917480468..	13.026779174804688	20.354342460632324
38	41	0	20.7	20349.4	24717.9	24966.6	27101.7	27329.0	64.0	77.00361	-13.00360870361328..	13.003608703613281	20.318138599395752
38	41	0	20.7	20429.5	24725.4	24982.1	27168.1	27352.8	64.0	76.96201	-12.96201324462890..	12.962013244628906	20.253145694732666
38	41	0	20.7	20592.2	24703.3	24961.7	27156.8	27351.4	64.0	76.74374	-12.74373626708984..	12.743736267089844	19.91208791732788
38	41	0	20.7	20629.8	24704.8	24964.0	27157.8	27352.0	64.0	76.72643	-12.72643280029296..	12.726432800292969	19.885051250457764

38	41	0	20.7	20514.8	24714.2	24971.8	27175.4	27368.9	64.0	76.650536	-12.65053558349609..	12.650535583496094	19.766461849212646
38	41	0	20.7	20550.6	24714.4	24971.8	27172.3	27366.2	64.0	76.650536	-12.65053558349609..	12.650535583496094	19.766461849212646
38	41	0	25.5	19063.6	25152.1	25404.9	27773.7	27949.4	79.0	90.85854	-11.85854339599609..	11.858543395996094	15.01081442531151
38	41	0	25.3	18937.0	25134.0	25389.1	27713.4	27886.4	79.0	90.4484	-11.44840240478515..	11.448402404785156	14.491648613652098
38	41	0	25.3	18931.5	25138.6	25393.2	27722.5	27893.9	79.0	90.4484	-11.44840240478515..	11.448402404785156	14.491648613652098
38	41	0	25.3	18924.7	25136.2	25392.3	27721.3	27893.1	79.0	90.4484	-11.44840240478515..	11.448402404785156	14.491648613652098
38	41	0	25.3	18910.2	25144.2	25400.9	27735.2	27885.0	79.0	90.441765	-11.44176483154296..	11.441764831542969	14.48324662220629
38	41	0	25.3	18911.7	25144.4	25403.3	27738.0	27893.7	79.0	90.441765	-11.44176483154296..	11.441764831542969	14.48324662220629
38	41	0	25.3	18920.3	25146.3	25401.3	27739.9	27895.0	79.0	90.441765	-11.44176483154296..	11.441764831542969	14.48324662220629
38	41	0	25.3	18917.1	25139.7	25394.3	27725.2	27898.9	79.0	90.426414	-11.42641448974609..	11.426414489746094	14.463815809805183
38	41	0	25.5	18940.4	25149.4	25404.0	27749.6	27908.2	79.0	90.41978	-11.41977691650390..	11.419776916503906	14.455413818359375
38	41	0	25.3	18913.1	25143.2	25389.7	27718.3	27912.8	79.0	90.41978	-11.41977691650390..	11.419776916503906	14.455413818359375

Registro Tecnico e Metadados

```
--- DADOS DA EXECUÇÃO ---
Data/Hora: 21/06/2026 13:51
Target: NivelPureza
Features: TeorRotulo, TeorDensimetro, NivelAcucar, Temperatura, CPJ_10k, CPJ_4k7, CPJ_3k19, CPJ_1k, CPJ_824R
Restrições: Livre
Estratégia: Personalizada via Aba de Configuração (Nested CV 12x3 p/ Otimização)
Seleção Topológica: Desabilitada
Amostras (Treino+CV): 1841 | Amostras (Cegas): 303
--- AVALIAÇÃO DE CONFIABILIDADE DOS DADOS ---
Volume: 1841 observações retidas.
Estabilidade Dimensional (CV Global): 19.26%
Complexidade Numérica (CN): 228697.35 [Alta/Crítica]
Colinearidade Original (VIF Médio): 430878.98
--- AVALIAÇÃO DE ALGORITMOS ---
> Modelo: Regressão Linear Múltipla
  Topologia: {'fit_intercept': True}
  Validação Interna: R² = 0.6050 | MAE = 5.2428 | RMSE = 6.6555 (6.97s)
  Validação Externa: R² = 0.5921 | MAE = 5.9216 | RMSE = 7.0513
  Análise de Resíduos: O teste de Breusch-Pagan rejeitou a hipótese de homocedasticidade (p=9.8668e-06).
  Interpretação: Os resíduos apresentam heterocedasticidade sistemática. Isto sugere que a variância do erro não é constante.
  Equação de Transferência: NivelPureza = (-123.314*TeorRotulo) + (-364.660*TeorDensimetro) + (-392.671*NivelAcucar) + (-1.123*Temperatura) + (-4.180*CPJ_10k) + (103.291*CPJ_4k7) + (-55.11
6*CPJ_3k19) + (-62.843*CPJ_1k) + (39.887*CPJ_824R) + 87.556
> Modelo: Regressão Bayesiana
  Topologia: {'alpha_1': 1.0, 'lambda_1': 1e-06}
  Validação Interna: R² = 0.6051 | MAE = 5.2441 | RMSE = 6.6553 (0.73s)
  Validação Externa: R² = 0.5930 | MAE = 5.9052 | RMSE = 7.0434
  Análise de Resíduos: O teste de Breusch-Pagan rejeitou a hipótese de homocedasticidade (p=8.2016e-06).
  Interpretação: Os resíduos apresentam heterocedasticidade sistemática. Isto sugere que a variância do erro não é constante.
  Equação de Transferência: NivelPureza = (-122.347*TeorRotulo) + (-361.698*TeorDensimetro) + (-389.514*NivelAcucar) + (-1.130*Temperatura) + (-4.278*CPJ_10k) + (102.538*CPJ_4k7) + (-54.72
4*CPJ_3k19) + (-62.434*CPJ_1k) + (39.546*CPJ_824R) + 87.556
> Modelo: SVR
  Topologia: {'C': 10, 'epsilon': 0.5, 'kernel': 'rbf'}
  Validação Interna: R² = 0.8340 | MAE = 3.0730 | RMSE = 4.3037 (8.79s)
  Validação Externa: R² = 0.4966 | MAE = 6.1049 | RMSE = 7.8338
  Análise de Resíduos: O teste de Breusch-Pagan rejeitou a hipótese de homocedasticidade (p=8.5179e-37).
  Interpretação: Os resíduos apresentam heterocedasticidade sistemática. Isto sugere que a variância do erro não é constante.
> Modelo: Random Forest
  Topologia: {'max_depth': 10, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 300}
  Validação Interna: R² = 0.9811 | MAE = 0.6689 | RMSE = 1.4495 (235.04s)
  Validação Externa: R² = 0.7341 | MAE = 3.4135 | RMSE = 5.6930
  Análise de Resíduos: O teste de Breusch-Pagan rejeitou a hipótese de homocedasticidade (p=3.3049e-21).
  Interpretação: Os resíduos apresentam heterocedasticidade sistemática. Isto sugere que a variância do erro não é constante.
> Modelo: Gradient Boosting
  Topologia: {'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 400, 'subsample': 0.8}
  Validação Interna: R² = 0.9866 | MAE = 0.5814 | RMSE = 1.2162 (101.62s)
  Validação Externa: R² = 0.8519 | MAE = 2.7084 | RMSE = 4.2494
  Análise de Resíduos: O teste de Breusch-Pagan rejeitou a hipótese de homocedasticidade (p=9.9725e-29).
  Interpretação: Os resíduos apresentam heterocedasticidade sistemática. Isto sugere que a variância do erro não é constante.
> Modelo: ANN
  Topologia: {'activation': 'tanh', 'alpha': 0.1, 'hidden_layer_sizes': (100, 100), 'solver': 'lbfgs'}
  Validação Interna: R² = 0.9941 | MAE = 0.3483 | RMSE = 0.7336 (1251.12s)
  Validação Externa: R² = 0.8002 | MAE = 2.8419 | RMSE = 4.9346
  Análise de Resíduos: O teste de Breusch-Pagan rejeitou a hipótese de homocedasticidade (p=6.2982e-25).
  Interpretação: Os resíduos apresentam heterocedasticidade sistemática. Isto sugere que a variância do erro não é constante.
> Modelo: XGBoost
  Topologia: {'colsample_bytree': 0.6, 'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 500, 'subsample': 0.7}
  Validação Interna: R² = 0.9746 | MAE = 0.8387 | RMSE = 1.6734 (53.56s)
```

```
  Validação Externa: R² = 0.8317 | MAE = 2.6612 | RMSE = 4.5301
  Análise de Resíduos: O teste de Breusch-Pagan rejeitou a hipótese de homocedasticidade (p=4.5261e-23).
  Interpretação: Os resíduos apresentam heterocedasticidade sistemática. Isto sugere que a variância do erro não é constante.
--- INFERÊNCIA ESTATÍSTICA (FRIEDMAN NO K-FOLD) ---
Metodologia: Comparação em Blocos (K-Fold CV)
Grupos Avaliados: Regressão Linear Múltipla, Regressão Bayesiana, SVR, Random Forest, Gradient Boosting, ANN, XGBoost
Estatística de Teste: 173.943 | Valor-p: 6.5539e-35
Hipótese Nula (H0): Os modelos possuem distribuições de erro estatisticamente equivalentes.
Decisão: Rejeita-se H0 (Existe modelo superior).
Tempo Computacional: 27m 38.0s
----------------------------------------
RAIO-X COMBINATÓRIO
--- Regressão Linear Múltipla ---
 * Tentativa {'fit_intercept': True} -> R² = 0.6141
 * Tentativa {'fit_intercept': False} -> R² = -66.7953
--- Regressão Bayesiana ---
 * Tentativa {'alpha_1': 1e-06, 'lambda_1': 1e-06} -> R² = 0.6141
 * Tentativa {'alpha_1': 1e-06, 'lambda_1': 0.0001} -> R² = 0.6141
 * Tentativa {'alpha_1': 1e-06, 'lambda_1': 1.0} -> R² = 0.6141
 * Tentativa {'alpha_1': 0.0001, 'lambda_1': 1e-06} -> R² = 0.6141
 * Tentativa {'alpha_1': 0.0001, 'lambda_1': 0.0001} -> R² = 0.6141
 * Tentativa {'alpha_1': 0.0001, 'lambda_1': 1.0} -> R² = 0.6141
 * Tentativa {'alpha_1': 1.0, 'lambda_1': 1e-06} -> R² = 0.6141
 * Tentativa {'alpha_1': 1.0, 'lambda_1': 0.0001} -> R² = 0.6141
 * Tentativa {'alpha_1': 1.0, 'lambda_1': 1.0} -> R² = 0.6141
--- SVR ---
 * Tentativa {'C': 0.1, 'epsilon': 0.01, 'kernel': 'rbf'} -> R² = 0.0912
 * Tentativa {'C': 0.1, 'epsilon': 0.01, 'kernel': 'linear'} -> R² = 0.2890
 * Tentativa {'C': 0.1, 'epsilon': 0.1, 'kernel': 'rbf'} -> R² = 0.0919
 * Tentativa {'C': 0.1, 'epsilon': 0.1, 'kernel': 'linear'} -> R² = 0.2896
 * Tentativa {'C': 0.1, 'epsilon': 0.5, 'kernel': 'rbf'} -> R² = 0.1008
 * Tentativa {'C': 0.1, 'epsilon': 0.5, 'kernel': 'linear'} -> R² = 0.2894
 * Tentativa {'C': 1, 'epsilon': 0.01, 'kernel': 'rbf'} -> R² = 0.5912
 * Tentativa {'C': 1, 'epsilon': 0.01, 'kernel': 'linear'} -> R² = 0.3709
 * Tentativa {'C': 1, 'epsilon': 0.1, 'kernel': 'rbf'} -> R² = 0.5919
 * Tentativa {'C': 1, 'epsilon': 0.1, 'kernel': 'linear'} -> R² = 0.3712
 * Tentativa {'C': 1, 'epsilon': 0.5, 'kernel': 'rbf'} -> R² = 0.5911
 * Tentativa {'C': 1, 'epsilon': 0.5, 'kernel': 'linear'} -> R² = 0.3705
 * Tentativa {'C': 10, 'epsilon': 0.01, 'kernel': 'rbf'} -> R² = 0.8202
 * Tentativa {'C': 10, 'epsilon': 0.01, 'kernel': 'linear'} -> R² = 0.4020
 * Tentativa {'C': 10, 'epsilon': 0.1, 'kernel': 'rbf'} -> R² = 0.8204
 * Tentativa {'C': 10, 'epsilon': 0.1, 'kernel': 'linear'} -> R² = 0.4024
 * Tentativa {'C': 10, 'epsilon': 0.5, 'kernel': 'rbf'} -> R² = 0.8209
 * Tentativa {'C': 10, 'epsilon': 0.5, 'kernel': 'linear'} -> R² = 0.4028
--- Random Forest ---
 * Tentativa {'max_depth': 5, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 200} -> R² = 0.7746
 * Tentativa {'max_depth': 5, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 300} -> R² = 0.7749
 * Tentativa {'max_depth': 5, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 500} -> R² = 0.7752
 * Tentativa {'max_depth': 5, 'min_samples_leaf': 1, 'min_samples_split': 10, 'n_estimators': 200} -> R² = 0.7734
 * Tentativa {'max_depth': 5, 'min_samples_leaf': 1, 'min_samples_split': 10, 'n_estimators': 300} -> R² = 0.7737
 * Tentativa {'max_depth': 5, 'min_samples_leaf': 1, 'min_samples_split': 10, 'n_estimators': 500} -> R² = 0.7740
 * Tentativa {'max_depth': 5, 'min_samples_leaf': 2, 'min_samples_split': 2, 'n_estimators': 200} -> R² = 0.7737
 * Tentativa {'max_depth': 5, 'min_samples_leaf': 2, 'min_samples_split': 2, 'n_estimators': 300} -> R² = 0.7740
 * Tentativa {'max_depth': 5, 'min_samples_leaf': 2, 'min_samples_split': 2, 'n_estimators': 500} -> R² = 0.7744
 * Tentativa {'max_depth': 5, 'min_samples_leaf': 2, 'min_samples_split': 10, 'n_estimators': 200} -> R² = 0.7727
 * Tentativa {'max_depth': 5, 'min_samples_leaf': 2, 'min_samples_split': 10, 'n_estimators': 300} -> R² = 0.7730
 * Tentativa {'max_depth': 5, 'min_samples_leaf': 2, 'min_samples_split': 10, 'n_estimators': 500} -> R² = 0.7734
 * Tentativa {'max_depth': 10, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 200} -> R² = 0.9763
 * Tentativa {'max_depth': 10, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 300} -> R² = 0.9764
 * Tentativa {'max_depth': 10, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 500} -> R² = 0.9764
```

```
 * Tentativa {'max_depth': 10, 'min_samples_leaf': 1, 'min_samples_split': 10, 'n_estimators': 200} -> R² = 0.9711
 * Tentativa {'max_depth': 10, 'min_samples_leaf': 1, 'min_samples_split': 10, 'n_estimators': 300} -> R² = 0.9712
 * Tentativa {'max_depth': 10, 'min_samples_leaf': 1, 'min_samples_split': 10, 'n_estimators': 500} -> R² = 0.9710
 * Tentativa {'max_depth': 10, 'min_samples_leaf': 2, 'min_samples_split': 2, 'n_estimators': 200} -> R² = 0.9723
 * Tentativa {'max_depth': 10, 'min_samples_leaf': 2, 'min_samples_split': 2, 'n_estimators': 300} -> R² = 0.9723
 * Tentativa {'max_depth': 10, 'min_samples_leaf': 2, 'min_samples_split': 2, 'n_estimators': 500} -> R² = 0.9724
 * Tentativa {'max_depth': 10, 'min_samples_leaf': 2, 'min_samples_split': 10, 'n_estimators': 200} -> R² = 0.9685
 * Tentativa {'max_depth': 10, 'min_samples_leaf': 2, 'min_samples_split': 10, 'n_estimators': 300} -> R² = 0.9686
 * Tentativa {'max_depth': 10, 'min_samples_leaf': 2, 'min_samples_split': 10, 'n_estimators': 500} -> R² = 0.9686
 * Tentativa {'max_depth': 8, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 200} -> R² = 0.9555
 * Tentativa {'max_depth': 8, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 300} -> R² = 0.9554
 * Tentativa {'max_depth': 8, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 500} -> R² = 0.9556
 * Tentativa {'max_depth': 8, 'min_samples_leaf': 1, 'min_samples_split': 10, 'n_estimators': 200} -> R² = 0.9513
 * Tentativa {'max_depth': 8, 'min_samples_leaf': 1, 'min_samples_split': 10, 'n_estimators': 300} -> R² = 0.9512
 * Tentativa {'max_depth': 8, 'min_samples_leaf': 1, 'min_samples_split': 10, 'n_estimators': 500} -> R² = 0.9514
 * Tentativa {'max_depth': 8, 'min_samples_leaf': 2, 'min_samples_split': 2, 'n_estimators': 200} -> R² = 0.9522
 * Tentativa {'max_depth': 8, 'min_samples_leaf': 2, 'min_samples_split': 2, 'n_estimators': 300} -> R² = 0.9520
 * Tentativa {'max_depth': 8, 'min_samples_leaf': 2, 'min_samples_split': 2, 'n_estimators': 500} -> R² = 0.9523
 * Tentativa {'max_depth': 8, 'min_samples_leaf': 2, 'min_samples_split': 10, 'n_estimators': 200} -> R² = 0.9489
 * Tentativa {'max_depth': 8, 'min_samples_leaf': 2, 'min_samples_split': 10, 'n_estimators': 300} -> R² = 0.9489
 * Tentativa {'max_depth': 8, 'min_samples_leaf': 2, 'min_samples_split': 10, 'n_estimators': 500} -> R² = 0.9492
--- Gradient Boosting ---
 * Tentativa {'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 200, 'subsample': 1.0} -> R² = 0.9585
 * Tentativa {'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 200, 'subsample': 0.8} -> R² = 0.9612
 * Tentativa {'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 400, 'subsample': 1.0} -> R² = 0.9710
 * Tentativa {'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 400, 'subsample': 0.8} -> R² = 0.9720
 * Tentativa {'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 200, 'subsample': 1.0} -> R² = 0.9817
 * Tentativa {'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 200, 'subsample': 0.8} -> R² = 0.9836
 * Tentativa {'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 400, 'subsample': 1.0} -> R² = 0.9827
 * Tentativa {'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 400, 'subsample': 0.8} -> R² = 0.9847
 * Tentativa {'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 200, 'subsample': 1.0} -> R² = 0.9269
 * Tentativa {'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 200, 'subsample': 0.8} -> R² = 0.9329
 * Tentativa {'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 400, 'subsample': 1.0} -> R² = 0.9580
 * Tentativa {'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 400, 'subsample': 0.8} -> R² = 0.9619
 * Tentativa {'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 200, 'subsample': 1.0} -> R² = 0.9765
 * Tentativa {'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 200, 'subsample': 0.8} -> R² = 0.9787
 * Tentativa {'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 400, 'subsample': 1.0} -> R² = 0.9819
 * Tentativa {'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 400, 'subsample': 0.8} -> R² = 0.9832
--- ANN ---
 * Tentativa {'activation': 'tanh', 'alpha': 0.1, 'hidden_layer_sizes': (50,), 'solver': 'lbfgs'} -> R² = 0.9847
 * Tentativa {'activation': 'tanh', 'alpha': 0.1, 'hidden_layer_sizes': (50, 50), 'solver': 'lbfgs'} -> R² = 0.9939
 * Tentativa {'activation': 'tanh', 'alpha': 0.1, 'hidden_layer_sizes': (100, 100), 'solver': 'lbfgs'} -> R² = 0.9948
 * Tentativa {'activation': 'tanh', 'alpha': 1, 'hidden_layer_sizes': (50,), 'solver': 'lbfgs'} -> R² = 0.9852
 * Tentativa {'activation': 'tanh', 'alpha': 1, 'hidden_layer_sizes': (50, 50), 'solver': 'lbfgs'} -> R² = 0.9926
 * Tentativa {'activation': 'tanh', 'alpha': 1, 'hidden_layer_sizes': (100, 100), 'solver': 'lbfgs'} -> R² = 0.9938
--- XGBoost ---
 * Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 200, 'subsample': 0.8} -> R² = 0.9027
 * Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 200, 'subsample': 1.0} -> R² = 0.8813
 * Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 200, 'subsample': 0.7} -> R² = 0.9030
 * Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 500, 'subsample': 0.8} -> R² = 0.9457
 * Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 500, 'subsample': 1.0} -> R² = 0.9186
 * Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 500, 'subsample': 0.7} -> R² = 0.9449
 * Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 200, 'subsample': 0.8} -> R² = 0.9630
 * Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 200, 'subsample': 1.0} -> R² = 0.9564
 * Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 200, 'subsample': 0.7} -> R² = 0.9628
 * Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 500, 'subsample': 0.8} -> R² = 0.9723
 * Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 500, 'subsample': 1.0} -> R² = 0.9672
 * Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 500, 'subsample': 0.7} -> R² = 0.9727
 * Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 200, 'subsample': 0.8} -> R² = 0.8649
 * Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 200, 'subsample': 1.0} -> R² = 0.8394
```

* Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 200, 'subsample': 0.7} -> R^2 = 0.8692
* Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 500, 'subsample': 0.8} -> R^2 = 0.9252
* Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 500, 'subsample': 1.0} -> R^2 = 0.8827
* Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 500, 'subsample': 0.7} -> R^2 = 0.9266
* Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 200, 'subsample': 0.8} -> R^2 = 0.9461
* Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 200, 'subsample': 1.0} -> R^2 = 0.9420
* Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 200, 'subsample': 0.7} -> R^2 = 0.9534
* Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 500, 'subsample': 0.8} -> R^2 = 0.9693
* Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 500, 'subsample': 1.0} -> R^2 = 0.9588
* Tentativa {'colsample_bytree': 1.0, 'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 500, 'subsample': 0.7} -> R^2 = 0.9714
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 200, 'subsample': 0.8} -> R^2 = 0.9045
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 200, 'subsample': 1.0} -> R^2 = 0.9037
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 200, 'subsample': 0.7} -> R^2 = 0.9048
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 500, 'subsample': 0.8} -> R^2 = 0.9458
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 500, 'subsample': 1.0} -> R^2 = 0.9380
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 500, 'subsample': 0.7} -> R^2 = 0.9434
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 200, 'subsample': 0.8} -> R^2 = 0.9650
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 200, 'subsample': 1.0} -> R^2 = 0.9600
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 200, 'subsample': 0.7} -> R^2 = 0.9679
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 500, 'subsample': 0.8} -> R^2 = 0.9727
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 500, 'subsample': 1.0} -> R^2 = 0.9712
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 500, 'subsample': 0.7} -> R^2 = 0.9770
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 200, 'subsample': 0.8} -> R^2 = 0.8644
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 200, 'subsample': 1.0} -> R^2 = 0.8509
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 200, 'subsample': 0.7} -> R^2 = 0.8672
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 500, 'subsample': 0.8} -> R^2 = 0.9246
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 500, 'subsample': 1.0} -> R^2 = 0.9127
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 500, 'subsample': 0.7} -> R^2 = 0.9295
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 200, 'subsample': 0.8} -> R^2 = 0.9475
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 200, 'subsample': 1.0} -> R^2 = 0.9309
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 200, 'subsample': 0.7} -> R^2 = 0.9472
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 500, 'subsample': 0.8} -> R^2 = 0.9699
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 500, 'subsample': 1.0} -> R^2 = 0.9621
* Tentativa {'colsample_bytree': 0.6, 'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 500, 'subsample': 0.7} -> R^2 = 0.9709